

Fountain Head Hybrid Energy Pre-DD

Development & Financeability Screen

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Project	Fountain Head hybrid energy desktop screen
Document type	Pre-DD Development & Financeability Screen
Basis	Public information, desktop map interpretation and indicative financeability screening
Status	Discussion draft - not for investment, financing, EPC or permitting decision-making
Core positioning	S4 60% renewable hybrid + A/B site hypothesis is the preferred pathway for next-step validation.

Important notice

This document is a public-information desktop screen. It is not an EPC design, engineering design, bankable feasibility study, lender due diligence report, legal advice, tax advice, environmental approval advice, investment recommendation or financing offer.

1. Executive Summary

This public version is intended for website publication, early client engagement and non-confidential discussion. It explains why Fountain Head-type remote mine loads are suitable for a structured PV+BESS + diesel hybrid Pre-DD screen, while keeping the analysis within a public-information boundary.

The current desktop conclusion is that S4 60% renewable hybrid + A/B site hypothesis is the most practical pathway for next-step validation. Zone A is treated as the BESS / PCS / EMS / interconnection node, Zone B as the main PV array candidate, and Zone C as a future expansion sensitivity only.

Core conclusion

The project should not be framed as a simple diesel-versus-100% renewable choice. The more useful Pre-DD question is which hybrid structure can reduce diesel exposure, preserve mine reliability, manage residual value and buy-out risk, and remain financeable for an owner, SPV and lender.

1.1 Current readiness snapshot

Framework	Theme	Current public-screen conclusion	Indicative maturity
F1	Approval Delta / Residual Approvals	The approval-risk categories have been identified, but the outcome depends on approved disturbance, MMP / licence, AAPA, water, fire, closure and ownership-chain confirmation.	6.5-7/10
F2	Site Suitability / Layout	A+B is the preferred desktop site hypothesis. A supports interconnection and BESS; B supports the main PV field; C is retained only as an expansion sensitivity.	7-7.5/10
F3	Connection & Microgrid Boundary	The project is best treated as a diesel-PV-BESS remote mine microgrid, not a standard grid-connected C&I solar-battery project.	5.5-6/10
F4	Commercial & Financial Close Readiness	S4 60% renewable hybrid is the preferred screening base case. S5 remains an upper-bound decarbonisation case, not the preferred commercial pathway.	7-7.5/10

1.2 What this report does and does not do

This report does	This report does not
Build a public-information development-readiness screen around F1-F4.	Confirm that the site is approved, buildable or bankable.
Identify a preferred desktop site and commercial hypothesis.	Replace owner data, engineering design, legal review, environmental approval review or lender due diligence.

This report does	This report does not
Translate public maps and screening finance into practical next-step questions.	Provide a final EPC layout, system design, tariff offer or investment recommendation.
Define what should be requested from the owner data room.	Resolve disturbance, heritage, fire, water, electrical or commercial boundaries.

2. Project Positioning and Report Basis

2.1 Why this case matters

Fountain Head-type projects represent a common challenge in remote mining energy: an onsite diesel power station supports a processing load, but the mine life, load horizon, residual asset value and technical integration pathway determine whether a renewable hybrid structure is commercially viable.

The purpose of the Pre-DD screen is to move from a broad opportunity statement to a structured development question: is there a technically plausible, site-aware, approval-aware and financeable hybrid pathway that is worth taking into owner data-room validation?

2.2 Public-information boundary

- Public materials can support a desktop hypothesis about project context, candidate zones, broad constraints and key data gaps.
- Public materials generally cannot close approved disturbance, final mine layout, SLD, cable route, hourly load, fire separation, AAPA coverage, closure obligations or bankability.
- The correct output is therefore a readiness screen and data request, not a final technical or financing conclusion.

2.3 Current strongest thesis

Layer	Preferred thesis	Why it matters
Commercial case	S4 60% renewable hybrid	Balances diesel reduction, owner economics, SPV financeability and integration risk better than a short-life standalone case or a 100% renewable screen.
Site case	A+B configuration	Zone A supports the electrical interconnection node; Zone B supports the main PV generation field.
Technical case	Diesel-PV-BESS microgrid	The project should be assessed as a remote mine microgrid with diesel as reliability backbone, not as a simple grid-connected battery project.
Next step	Owner data room + hourly dispatch + EPC pricing + commercial term sheet	This is the practical bridge from public screen to specialist DD and financial-close preparation.

3. F1-F4 Pre-DD Methodology

The report uses a four-framework structure. Each framework answers a different early-stage development question and produces a different class of data request.

Framework	Core question	Primary risk if unresolved	Main evidence required
F1 - Approval Delta	Can the project be permitted or accommodated within the existing approval pathway?	The project may require approval variation, additional studies, delayed CP satisfaction or changed ownership arrangements.	Approved disturbance footprint, MMP / licence, AAPA, fire, water, closure and ownership-chain evidence.
F2 - Site Suitability	Is there a real, low-conflict, buildable and explainable site hypothesis?	The PV/BESS concept may be visually plausible but constrained by water, IWL, closure, roads, pipeline, slope or net area.	Latest layout, GIS, net usable area, slope / drainage, cable corridor, closure and surface-condition review.
F3 - Microgrid Boundary	Can PV, BESS, diesel and mine load be electrically integrated without compromising reliability?	Financeability and performance guarantees may fail if dispatch, protection, load data or control boundaries are not understood.	SLD, switchboard, gensets, minimum loading, hourly / sub-hourly load, EMS / PCS, protection and metering data.
F4 - Commercial Readiness	Can the project be structured so the owner saves money, the SPV earns a return and lenders can size debt?	The project may look attractive in a spreadsheet but fail on buy-out, mine life, diesel risk, EPC cost or lender CPs.	Demand horizon, tariff structure, SPV boundary, buy-out, EPC pricing, lender model and term-sheet assumptions.

Practical memory aid

F1 asks: can it be approved? F2 asks: can it be placed? F3 asks: can it be connected and operated reliably? F4 asks: can it be signed and financed?

4. F1 - Approval Delta / Residual Approvals

F1 tests whether the PV+BESS concept creates a new approval delta relative to the existing mine approval pathway, disturbance footprint, water management, fire and hazardous materials environment, closure obligations, cultural heritage framework and legal ownership chain.

4.1 Ten F1 questions

No.	Question	Current desktop view	Open evidence
1	Existing approval baseline	Public material supports an existing mining and power-infrastructure context, but not the	Latest MMP, environmental approval conditions and power-infrastructure scope.

No.	Question	Current desktop view	Open evidence
		exact PV+BESS approval position.	
2	Disturbance footprint	A+B appears directionally close to infrastructure and disturbed land, but the approved disturbance footprint is not confirmed.	Approved disturbance GIS polygon and construction footprint.
3	Licence variation / amendment	PV+BESS may be characterised as supporting energy infrastructure, but this cannot be assumed.	Regulator view, licence conditions, MMP amendment requirements.
4	Water / stormwater	Watercourse, pond, pit and drainage constraints are material for B, A-B corridor and BESS firewater.	Water management plan, flood / drainage paths, sediment and firewater containment.
5	Fire / hazardous materials	Zone A is attractive for interconnection but may be close to fuel farm, plant and power-station hazards.	Fire separation, fuel-farm layout, BESS emergency-response requirements, insurer input.
6	Closure / rehabilitation	Mine life may be shorter than PV/BESS asset life, creating residual-value and removal obligations.	Closure plan, rehabilitation assumptions, buy-out / relocation / decommissioning mechanism.
7	Cultural heritage	New ground disturbance, pile driving and cable trenching may require coverage confirmation.	AAPA Authority Certificate coverage and exclusion areas.
8	Biodiversity / EPBC	New clearing or proximity to watercourse / habitat could create ecological approval questions.	PMST / ecology screen, clearing footprint, watercourse buffers.
9	Approval-holder chain	A third-party SPV structure requires alignment between approval holder, operator, site rights and counterparty.	Entity chart, site licence / lease, direct agreement and step-in rights.
10	Timing / CP mapping	The key issue is not only what must be done, but when it must be done.	FC, NTP, COD and operating-period CP register.

4.2 F1 conclusion

A+B is a reasonable public desktop approval-delta hypothesis, but it cannot be presented as approved or approval-ready. The key approval questions sit around disturbance, water, fire, AAPA / heritage, ecology, closure, SPV rights and CP timing.

F1 headline

Zone A is mainly an approval, fire and access question. Zone B is mainly a disturbance, water, ecology, heritage and closure question. The A-B cable corridor connects both risk profiles.

5. F2 - Site Suitability / Layout

F2 tests whether the project has a credible physical site hypothesis. The question is not simply whether land appears open on a map, but whether a PV+BESS system can be placed in a way that is spatially, operationally, environmentally and commercially defensible.

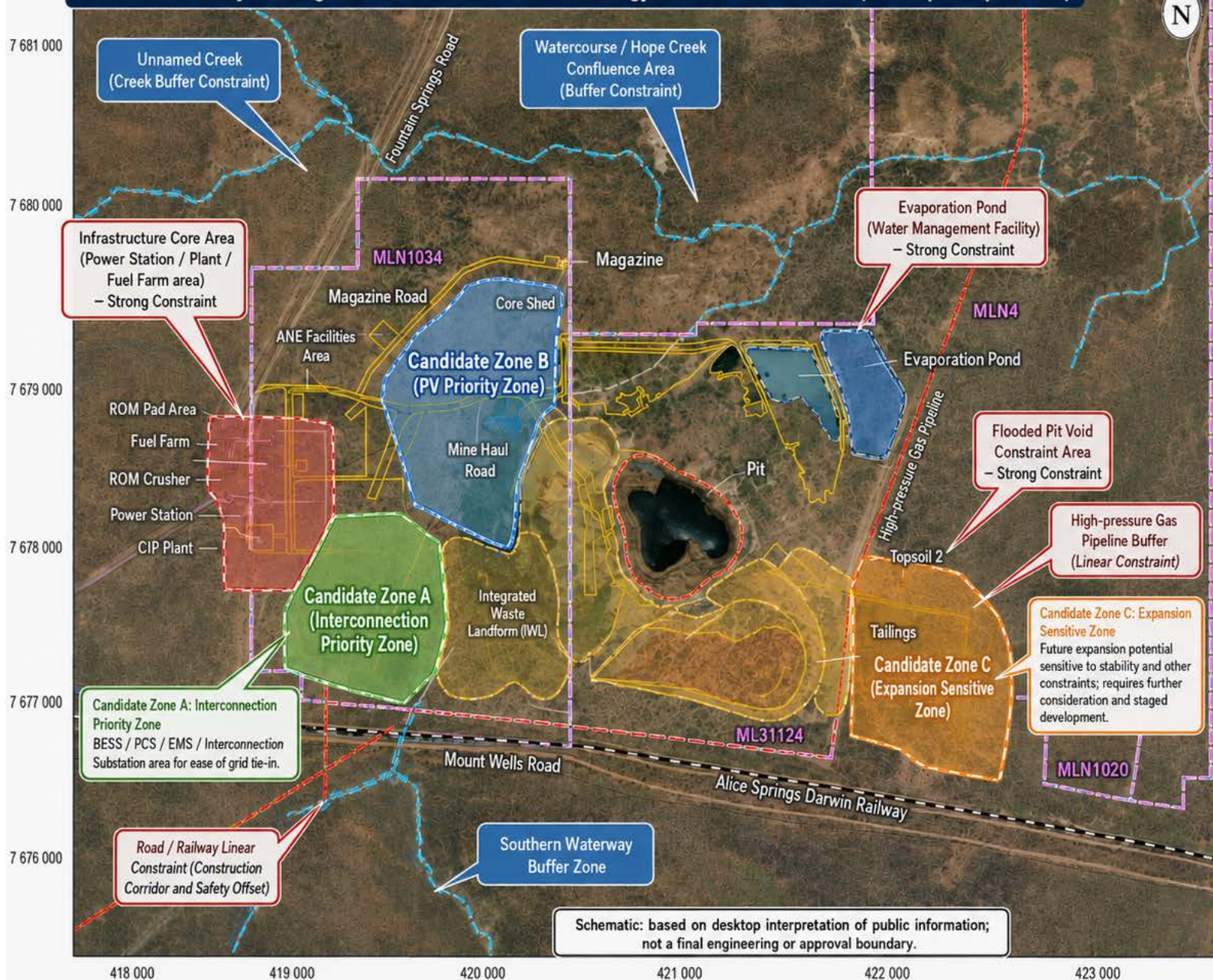
5.1 Candidate zone logic

Zone	Preferred function	Public desktop rationale	Critical unresolved items
A	Interconnection priority zone - BESS / PCS / EMS / switchgear / interconnection	Close to the power station, processing infrastructure and likely electrical node. Best suited to control and interconnection equipment rather than a large PV field.	Fire separation, fuel interface, plant operations, stormwater / firewater containment, access, switchboard proximity and available pad.
B	PV priority zone - main PV array candidate	Appears to provide better spatial potential for a PV field and is more plausible than A for a 10-13 MWp scale array in the S4 screening case.	Approved disturbance, net usable area, slope, drainage, IWL / waste / closure interface, access and cable corridor to A.
C	Expansion sensitivity only	May be relevant if the owner confirms future hub demand or expansion-side load. It is not the preferred base case.	Tailings / topsoil / closure interface, pipeline and railway constraints, longer cable route, higher approval delta and unclear long-term function.

Site logic

A is where the system connects. B is where the main PV field is most plausibly placed. C is not the base case; it remains a future expansion sensitivity.

Fountain Head Project: Integrated Schematic of Candidate Energy Zones and Constraints (Desktop Interpretation)



Legend

Candidate Energy Zones (Preferred Direction)

- Candidate Zone A: Interconnection Priority Zone BESS / PCS / EMS / Interconnection
- Candidate Zone B: PV Priority Zone Solar PV array
- Candidate Zone C: Expansion Sensitive Zone Future expansion area

Constraint Categories and Types

Strong Constraints (Red Family)

- Infrastructure Core Area (Power Station / Plant / Fuel Farm area)
- Flooded Pit Void
- Water Management Facility (Evaporation Pond)
- Natural Gas Pipeline Buffer (High-pressure Pipeline)

Medium Constraints / Needs Review (Yellow Family)

- IWL / Waste Landform
- Topsoil / Rehabilitation Area (Topsoil / Tailings Closure)
- Access Roads / Tracks

Waterbody / Watercourse Constraints (Blue Family)

- Fountain Head Lake Constraint Area
- Creeks / Watercourse (Including Unnamed Creek and Other Tributaries)

Other Symbols

- Mining Lease Boundary
- Infrastructure / Roads
- Railway
- Road
- Gas Pipeline (High Pressure)
- Watercourse / Waterbody (Existing)

5.2 F2 current answer

The current F2 answer is that A+B is the preferred desktop site hypothesis for the S4 case. Zone A is the preferred location for BESS, PCS, EMS and interconnection equipment. Zone B is the preferred main PV array candidate. Zone C should not be used as the base-case site unless the owner confirms a future hub or expansion-side demand centre.

5.3 What remains open

- Whether A+B is fully within the approved disturbance footprint.
- Net usable area for the main PV array after excluding water, roads, closure areas, IWL, tailings, heritage exclusions and buffers.
- Slope, surface condition and geotechnical suitability of Zone B.
- BESS pad availability, fire separation and stormwater / firewater containment in Zone A.
- A-B cable route, including road, rail, watercourse and pipeline crossing risks.
- Latest owner mine layout, which may supersede public EIS-style mapping.

6. F3 - Connection & Microgrid Boundary

F3 tests whether the site hypothesis can become a reliable electrical system. For this type of project, the relevant question is not simply where the grid connection is. The relevant question is how PV, BESS, diesel generation, processing load, control systems, metering and ownership boundaries work together.

6.1 Preferred microgrid hypothesis

Microgrid layer	Current view	Why it matters
Existing diesel power station	Reliability backbone and current source of mine power.	Diesel minimum loading, spinning reserve and start / stop constraints affect renewable penetration.
PV generation field	Zone B is the preferred public desktop location for the main PV array.	PV cannot be assessed only on annual generation; it must be dispatchable within the diesel microgrid context.
BESS / PCS / EMS	Zone A is the preferred interconnection and control node.	BESS capability, PCS mode and EMS control determine whether higher renewable penetration is operationally feasible.
A-B cable corridor	A required but unconfirmed connection between PV field and interconnection node.	Crossings, trenching, voltage level, communications and approvals can materially affect cost and risk.
Diesel ownership boundary	Open issue.	Determines whether the SPV sells PV+BESS output, an availability service or a full hybrid power station outcome.

6.2 Critical F3 data requests

- Existing electrical single-line diagram (SLD).
- Main switchboard location, voltage level and spare feeder / breaker capacity.

- Diesel generator make, model, controller, fleet controller and operating philosophy.
- Minimum stable loading, ramp rates, start / stop constraints and spinning reserve requirements.
- Hourly and sub-hourly load data, including seasonal and operating-mode variation.
- Critical and non-critical load register, including feeder separation and load-shedding capability.
- Existing SCADA / PLC / DCS / power-station control architecture.
- Protection settings, fault-level information and planned protection study requirements.
- BESS / PCS requirements: grid-following vs grid-forming, black start, frequency support and fault current.
- Metering locations, data historian access and PPA / EaaS delivery point.
- A-B cable route options and crossing constraints.
- Ownership and control boundary: owner-retained diesel versus SPV-operated hybrid power station.

6.3 F3 conclusion

Public information supports treating the project as a diesel-PV-BESS remote mine microgrid opportunity. However, the microgrid boundary cannot be confirmed without SLD, diesel generator data, main switchboard data, hourly / sub-hourly load, EMS / PCS design, protection review, metering data, cable route and ownership / operating boundary confirmation.

7. F4 - Commercial & Financial Close Readiness

F4 tests whether the technical and site hypothesis can become a commercial structure that the owner can sign, the SPV can invest in and lenders can support. It is the bridge between the development screen and a potential commercial term sheet.

7.1 Scenario logic

Scenario	Current Pre-DD position	Commercial implication
S1 - Short-life standalone case	Not the preferred third-party SPV pathway.	Mine-life mismatch can create high buy-out / residual-value pressure.
S3 - Expanded hub, same sizing	Improves the demand horizon but may still leave owner economics weak.	Useful as a transition case, not the preferred reference case.
S4 - 60% renewable hybrid	Preferred screening base case.	Best current balance between owner savings, SPV financeability and practical renewable penetration.
S5 - 100% renewable screen	Upper-bound decarbonisation case only.	Higher CAPEX, integration risk and buy-out pressure may undermine owner economics.

7.2 Three-party commercial test

Party	Key question	Why S4 is the current reference case
Mine owner	Does the project deliver lifecycle savings without creating unacceptable reliability, closure or buy-out risk?	S4 is less capital-heavy than S5 and more decarbonising than a low-renewable short-life case.

Party	Key question	Why S4 is the current reference case
SPV / developer	Can project cash flow support return of and return on capital over the contract window?	S4 has a more credible balance of CAPEX, contracted revenue and residual-value exposure.
Lender	Can debt be sized against stable contracted cash flow with risks allocated through CPs and documentation?	S4 is more likely than S5 to pass practical debt sizing once technical and commercial gaps are closed.

7.3 F4 unresolved items

- Owner confirmation of mine life, processing horizon and hub demand.
- Whether diesel generators remain owner-owned or form part of an SPV-operated hybrid power station.
- Diesel fuel price risk allocation and dispatch-control responsibility.
- EaaS / PPA / BOO / owner-owned structure selection.
- Buy-out mechanism: fixed, formula-based, optional or mandatory.
- Early termination, mine closure, processing delay and extension rights.
- Carbon / ESG benefits allocation.
- Minimum offtake or availability payment.
- EPC and vendor pricing, including civil, cable, switchgear, augmentation and O&M.
- Hourly dispatch validation and lender-grade financial model.
- Direct agreement, lender step-in rights and financial-close CP register.

F4 headline

F4 has answered which commercial pathway is worth validating: S4. It has not yet answered exactly how S4 should be signed, who owns which assets, who controls diesel dispatch, who carries fuel risk or how lenders size debt.

8. Integrated Gap Register

The following gaps are ranked by importance for moving from public desktop screen to owner data-room validation and specialist DD.

Rank	Gap	Frameworks	Importance	Why it matters
1	Approved disturbance footprint	F1 / F2	10/10	Determines whether A+B is inside the already approved disturbance area or creates a new approval delta.
2	Latest mine layout / CAD / GIS	F1 / F2	9.5/10	Public maps may lag the current mine plan, road layout, power station position, tailings plan or closure layout.
3	Existing SLD and main switchboard data	F3	9.5/10	Defines where and whether BESS / PCS / PV can physically and safely interconnect.
4	Hourly and sub-hourly load data	F3 / F4	9/10	Annual energy is insufficient for dispatch, reliability, curtailment and debt sizing.
5	Diesel genset configuration and minimum load	F3	9/10	Diesel minimum loading and reserve requirements determine achievable renewable penetration.
6	Fire / BESS separation and fuel-farm layout	F1 / F2 / F3	9/10	Zone A is attractive but may be constrained by hazardous-area, fire and insurer requirements.
7	AAPA / cultural heritage coverage	F1 / F2	8.5/10	Ground disturbance, pile driving and trenching may require certificate coverage confirmation.
8	Water, drainage, flood and firewater containment	F1 / F2	8.5/10	B and A-B corridor interact with watercourses, pit water, ponds and wet-season risk.
9	Closure, rehabilitation and buy-out mechanism	F1 / F4	8.5/10	Asset life can exceed mine life, creating residual value, removal and rehabilitation obligations.
10	SPV / owner / approval-holder chain	F1 / F4	8.5/10	The legal ability for a third-party SPV to own, access, operate and finance assets must be established.
11	EPC / vendor pricing	F4	8/10	Screening CAPEX must be replaced by vendor-specific pricing before term sheet and debt sizing.
12	A-B cable route	F2 / F3	8/10	The cable corridor can create crossing, trenching, cost, disturbance

Rank	Gap	Frameworks	Importance	Why it matters
				and heritage issues.
13	Lender-grade model	F4	8/10	Debt sculpting, reserves, downside cases and covenants are required before bankability conclusions.
14	DEM / QGIS / slope screen	F2	6.5-7/10	Useful site-confidence enhancement, but not the highest-priority blocker without owner GIS.

9. Owner Data Request

The following request list is designed to convert the public-information screen into a structured owner conversation and data-room review.

Category	Requested data	Frameworks served
Approvals and land	Approved disturbance footprint, latest MMP / deemed licence, environmental approval conditions, mining lease / tenure data.	F1 / F2
Site layout	Latest mine layout, CAD / GIS, power station, fuel farm, processing plant, haul roads, water infrastructure, tailings and closure areas.	F1 / F2
Heritage and environment	AAPA coverage, exclusion areas, ecology constraints, watercourse buffers, water management plan and closure plan.	F1 / F2
Electrical system	SLD, main switchboard, voltage levels, spare feeders, protection settings, metering and data historian access.	F3
Diesel system	Genset models, controllers, minimum load, ramp rates, spinning reserve and maintenance / operating philosophy.	F3
Load data	Hourly load, sub-hourly load, processing schedule, critical / non-critical loads and shutdown periods.	F3 / F4
Commercial boundary	Preferred ownership structure, diesel ownership, EaaS / PPA / BOO preference, buy-out position and willingness for minimum offtake.	F4
Finance and procurement	EPC pricing inputs, vendor preferences, O&M assumptions, insurance, tax, carbon / ESG treatment and financing constraints.	F4

10. Suggested First Meeting Agenda

Agenda item	Purpose	Owner decision or response sought
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Agenda item	Purpose	Owner decision or response sought
Confirm project and energy objective	Clarify whether the owner is solving for lowest cost, diesel reduction, ESG, power reliability or financeability.	Confirm priority order and constraints.
Test S4 as the reference case	Use S4 60% renewable hybrid as the working case for discussion.	Agree whether S4 is the right starting point for owner data-room validation.
Review A+B site hypothesis	Discuss whether A+B aligns with current mine layout and operations.	Confirm whether A+B is worth checking against approved disturbance and current layout.
Clarify microgrid boundary	Determine whether PV+BESS only or full hybrid power station service is preferred.	Identify diesel ownership, dispatch control and metering boundary.
Discuss commercial structure	Explore EaaS, PPA, BOO, owner-owned and buy-out options.	Identify acceptable risk allocation and term-sheet direction.
Agree next-step data request	Convert the public screen into a focused owner data request.	Confirm priority data, access process and next workshop.

11. CP Map - What Closes When

A useful output of the F1-F4 screen is to separate matters that must be closed before financial close from those that can be resolved before notice to proceed, before COD or during operations.

Stage	Typical items to close	Business meaning
Before Financial Close	Approved disturbance footprint, site rights, owner counterparty, SPV / owner boundary, SLD feasibility, demand horizon, tariff structure, lender security package.	Bankability and contractual readiness.
Before NTP	Detailed EPC design, final layout, construction method, AAPA / heritage work packs, fire strategy, drainage and erosion controls, outage planning.	Construction readiness.
Before COD	Commissioning tests, protection settings, EMS tuning, metering, operating procedures, handover, performance testing.	Operational readiness and revenue start.
During Operations	Availability reporting, performance monitoring, curtailment treatment, major maintenance, augmentation planning, extension / buy-out management.	Long-term performance and risk management.

12. Recommended Next-Step Pathway

1. Confirm mine life, processing horizon and potential hub-demand case.

2. Request owner data room using the F1-F4 evidence list.
3. Confirm whether A+B sits within approved disturbance and aligns with current mine layout.
4. Build an hourly dispatch model for S4 and test curtailment, diesel minimum load and reliability.
5. Prepare a microgrid concept note covering SLD interface, EMS / PCS approach, metering and cable route.
6. Obtain EPC / vendor pricing for PV, BESS, PCS, transformer, switchgear, cable, civil and O&M scope.
7. Compare EaaS, PPA, BOO and owner-owned structures with buy-out and early-termination treatment.
8. Build a lender-grade model with downside cases, debt sizing, reserves and CP tracking.
9. Decide whether the project is ready for specialist DD, term-sheet discussion and financial-close preparation.

Appendix A - F1-F4 Question Bank

Framework	Question bank summary
F1	Existing approval baseline; approved disturbance; licence variation; water / stormwater; fire / hazmat; closure / rehabilitation; cultural heritage; biodiversity / EPBC; approval-holder chain; CP timing.
F2	Candidate-zone logic; net usable area; slope / surface; drainage; IWL / waste / closure interface; access; cable corridor; latest mine layout; haul road / pipeline / railway crossings.
F3	SLD; switchboard; voltage level; gensets; minimum load; spinning reserve; hourly / sub-hourly load; EMS; PCS mode; protection; metering; ownership and control boundary.
F4	Demand horizon; SPV boundary; diesel ownership; diesel fuel risk; tariff / EaaS terms; buy-out; early termination; carbon / ESG allocation; EPC pricing; lender model; CP register.

Appendix B - Public Source Register

- NT EPA public register - Fountain Head Gold Project and related environmental assessment documentation.
- Katestone / ERIAS, Fountain Head Gold Project Air Quality and Greenhouse Gas Assessment, 30 March 2021.
- Northern Territory Resourcing the Territory - Developing Projects and Mining Developments reporting.
- Patronus Resources ASX announcement, Key Mining Approval Received for Fountain Head Gold Project, 4 November 2024.
- Public map extracts and desktop GIS-style interpretation prepared for Pre-DD discussion.
- Remote mine hybrid power references and public industry materials used for high-level microgrid framing.

Appendix C - Limitations

- This report is a public-information desktop screen and must not be used as an engineering, legal, environmental, financing, tax or investment decision document.
- A/B/C zones are candidate site hypotheses only. They are not surveyed development boundaries, approval boundaries or EPC layouts.
- The map interpretation is schematic and does not replace owner GIS, survey, DEM / slope analysis, geotechnical review, drainage design or fire engineering.
- S4 and S5 are indicative screening cases only and do not represent financing commitments, commercial offers, tariff proposals or investment advice.
- All conclusions require owner data-room review and specialist DD before any permitting, contracting, procurement, financing or construction decision.

Final conclusion

Public enhanced conclusion

S4 60% renewable hybrid is the preferred next commercial base case, and A+B is the preferred desktop site hypothesis. The project should now be advanced through owner data-room validation, hourly dispatch modelling, EPC pricing, approval-pathway confirmation and commercial term-sheet development. This is a screen, not a bankable conclusion.